

CalSoft vs Fluke and Beamex, A Capability Comparison

Repository: cirqen-labs | Branch: improvement-plan | Reviewed: 17 September 2026 Third in the series, after A1_ENGINEERING_AUDIT.pdf (formula audit) and 02_RATING_AND_COMPARISON.pdf (internal rating)

Revised 18 September 2026: six matrix rows changed after implementation.

0. Two notes before the comparison

On "Beamer". Read here as **Beamex**, the Finnish calibration company whose CMX calibration management software and MC6 field calibrators are the natural comparator alongside Fluke. If something else was meant (Beamer is also a LaTeX presentation class, and BEAMER an e-beam lithography tool, neither relevant to calibration), say so and this section can be redone.

On scope. "CalSoft" here means the calibration capability of the whole platform, which spans two git repositories: the cirqen-labs Django application, and a separate HQ server (~/Desktop/hq_server) that allocates certificate numbers and acts as the sync authority. Rows in the matrix below credit the platform for what either provides.

On comparing unlike things. The three names cover *three different product classes*, and collapsing them into one league table would produce a misleading answer. The single most useful result of this comparison is the recognition that CalSoft is not competing with one product, it sits between two:

Product	Class	What it is for
Fluke Biomedical Ansur	Hospital biomedical QA automation	Driving analyzers/simulators through electrical-safety and performance test protocols on medical devices
Fluke MET/CAL + MET/TEAM	Accredited metrology laboratory	Executing calibration procedures with a full GUM uncertainty budget and managing the asset estate
Beamex CMX (+ MC6)	Process/industrial calibration management	Managing calibration of process instrumentation end to end, with uncertainty and regulatory records
CalSoft	In-house hospital calibration and certification	Capturing manual readings, reducing them to an uncertainty budget, and issuing a numbered certificate

CalSoft's functional peer is **Ansur** (same setting, same devices, same users). Its *ambition*, a GUM uncertainty budget feeding a conformity statement, is

MET/CAL and CMX territory. That split is the whole story, and section 3 is the punchline.

1. Capability matrix

Sourced to vendor documentation (section 8). "Not documented" means the vendor's own literature does not describe the capability, not that it is impossible.

Capability	CalSoft	Ansur	MET/CAL + MET/TEAM	Beamex CMX	
Reading capture	Manual keyboard entry	Automated from analyzer via per-model plug-in	Automated, procedure drives the instruments	Automated, MC6 field calibrator syncs to CMX	
Type A (repeatability)	Yes, $s/\sqrt{n}$	Not documented	Yes (GUM since v6.0, 1999)	Yes	
Type B components	Two, fixed: resolution + reference	Not documented	User-defined, configurable	Up to eight user-defined	
Combined / expanded U	Yes, RSS then k	u_c	Not documented	Yes, GUM-compliant	Yes, per calibration point
U in the verdict	Yes , guard banding applied	n/a (limit-based)	Yes , guard banding adjusts limits during execution	Yes , flags "passed, but at risk of fail once uncertainty is counted"	
Decision outcomes	Pass / Fail / Indeterminate	Pass / Fail vs test limits	Pass / Fail / Indeterminate	Pass / Fail + documented risk view	
TUR ratio	Yes , printed per point, named warning below 4:1	Not documented	Yes	Yes, built into the uncertainty view	
Guard banding	Yes , acceptance and rejection limits from U	Not documented	Yes, dynamic	Risk-based equivalent	
Degrees of freedom	Yes , Welch-Satterthwaite; k derived, not assumed	Not documented	Part of GUM handling	Not separately documented	
Electronic signature	Pasted signature image (base64 PNG)	Real e-signature, 21 CFR Part 11	Yes, 21 CFR Part 11	Yes, 21 CFR Part 11 e-records/e-signatures	
Audit trail on records	App-level audit-log model, plus an HQ audit_log table	Version-controlled test records	All changes retained as audit trail	Full database change tracking	
Central certificate authority	Yes, HQ allocates numbers under an advisory lock	Central PC / CMMS records	MET/TEAM over MET/CAL	CMX server	
Certificate integrity	Numbers now single-authority; QR check still always returns valid	Version-controlled electronic records	Audit trail + e-sig	Permanent auditable store	
Accreditation	None claimed	ISO 9001 / 13485 manufacturer; supports hospital QA regimes	Supports ISO/IEC 17025, ANSI Z540, 21 CFR 11	Supports ISO 9001, ISO 17025, GMP/GAMP, 21 CFR 11	

Capability	CalSoft	Ansur	MET/CAL + MET/TEAM	Beamex CMX
Intervals	Yes, via calSchedules	PM justification from trend data	MET/TRACK: days, weeks, months, use cycles	Yes
Offline operation	Yes, CDC sync engine, 17k LOC	Central PC or standalone install	Central SQL Server	CMX server; MC6 works offline in the field
Batch testing	One session per device	Yes, multiple devices at once	Procedure-driven	Work-order driven
Licence cost	None (in-house)	Per analyzer plug-in, per computer	Substantial; needs MS SQL Server	Commercial licence
Platform	Django, PostgreSQL, Linux	Windows + MS SQL Server + .NET	Windows + MS SQL Server	Windows/server

2. Where CalSoft already holds its own

Three of these are genuine, and one is a real competitive advantage.

It computes an uncertainty budget where its functional peer does not. The Ansur technical datasheet describes the product as automatically assessing "pass/fail against test limits specified by global standards or organizations." Type A, Type B, combined and expanded uncertainty, coverage factors, guard banding and TUR appear nowhere in it. Ansur is a *test-limit* tool. CalSoft attempts the GUM budget that Ansur does not, and the five core formulas are individually correct (per A1_ENGINEERING_AUDIT.pdf section 3). **On the mathematics of uncertainty, CalSoft is ahead of the Fluke product actually sold for hospital biomedical QA**, and behind the Fluke product sold for metrology labs.

Offline-first operation is a real advantage, not a compromise. Ansur wants a central PC or a networked MS SQL Server; MET/CAL and MET/TEAM want SQL Server; CMX is a licensed server product. CalSoft runs a 17,000-line change-data-capture sync engine (sync/, Debezium table registration in sync/config.py) that lets a workshop calibrate through a network outage and reconcile afterwards. For a hospital estate where connectivity is intermittent, this is worth more than several of the features in the matrix above, and none of the three commercial products is architected for it.

Cost and fit. No per-analyzer plug-in licence, no per-computer licence, no SQL Server, and the data model already knows about the hospital's departments, workshops, roles and schedules. Ansur charges per analyzer plug-in per computer.

Procedure and schedule integration. CalSoft's procedures, parameters, set values, sub-parameters, equipment mappings and calibration schedules are one system. Reaching that with Fluke means MET/CAL *plus* MET/TEAM or MET/TRACK.

There is a real central certificate authority. This was understated in the first draft of this comparison. A separate HQ server allocates every certificate number under a Postgres advisory lock, writes its own audit_log, runs per-device API keys with issuance and revocation, and has a dedicated test suite for the numbering itself. Structurally that is closer to MET/TEAM's role over MET/CAL than a single Django application would suggest, and it is more than Ansur offers, which records to a central PC or CMMS without acting as an allocating authority.

The qualification is that the authority is **not exclusive**: the field application can also allocate, by a different algorithm, on a different database. Commercial platforms have one allocator by construction. That is the gap, not the absence of an authority, but the presence of two.

3. The one finding that matters

Put the matrix rows in the right order and the picture is unambiguous.

This section described the position on 17 September. It has changed, and the change is recorded at the end.

CalSoft had built **the hard part**, a GUM-shaped uncertainty budget, which Ansur does not attempt, and then omitted **the easy part that makes it mean something**: using the number in the decision.

- MET/CAL takes the uncertainty and *moves the test limits with it*, yielding Pass, Fail, or **Indeterminate** when a reading sits too close to the limit to call.
- Beamex CMX takes the uncertainty and shows the calibrator that a point "may be passed but when taking the uncertainty into account then there is a risk that it is failed," and checks the reference standard is good enough via TUR.
- CalSoft computes u_A , u_{res} , u_{ref} , u_c and U to six decimal places, prints them on the certificate, and then decides conformity with `abs(error) <= tolerance (models.py:680)`, uncertainty excluded.

The asymmetry is sharper still: **CalSoft already contains both missing pieces**.

`MetrologyUtils.calculate_guard_banding` and `calculate_measurement_capability` (TUR) are written, correct in outline, and have **no call sites anywhere in the repository**. The module's own 6,008-line handbook works a TUR example by hand in section 5.9.2 and discusses the guard-band u_c -versus- U ambiguity in section 5.10, concluding the function "must not be wired in without a decision being taken and documented."

So the distance from CalSoft to MET/CAL-class decision behaviour was not a development programme. It was **one decision and one function call**.

That distance has since been closed. Guard banding is applied, the decision rule is printed as ISO/IEC 17025 requires, the test uncertainty ratio appears per test point with a named warning below 4:1, and the coverage factor is derived from the effective degrees of freedom rather than assumed to be 2. On the six matrix rows covering decision behaviour, CalSoft now reads the same as MET/CAL and CMX.

Two qualifications matter. Commercial guard banding is configurable per laboratory and per capability; CalSoft applies one rule. And MET/CAL has done this since 1999, which is the more useful way to read the gap: the arithmetic was never the hard part, and being late to use it is the finding, not being unable to.

The second finding is less flattering and cannot be closed by a function call:

certificate integrity. Ansur, MET/TEAM and CMX all implement 21 CFR Part 11 electronic signatures with audit trails over the records. CalSoft pastes a base64 PNG of a handwritten signature into a PDF and prints "Certificate authenticity can be verified by scanning the QR code" above a verifier that returns `valid: True` unconditionally (`verification.py:195`). Against any of the three. This is the widest real gap, and the pattern is already in-house: the platform runs Ed25519 signing for update manifests (`cirqen-labs/hq_server/build_package.py:53`, the update server), and the HQ certificate server already runs per-device API keys with hashing and revocation.

Signing belongs at HQ rather than in CalSoft, because HQ is what allocates the number, and a signature is only as trustworthy as the identity it binds. That places it alongside the numbering question above: a signed certificate whose number might later be allocated to another device is not a trustworthy artefact. `04_PLAN_TO_9_OF_10.pdf` section 6 treats the two as one phase.

4. Where the gap is real, and what specifically to borrow

Gap	Borrow from	Concrete change
Uncertainty excluded from the verdict	MET/CAL guard banding; CMX risk view	Call the existing <code>calculate_guard_banding</code> ; add a third outcome
Binary Pass/Fail hides marginal results	MET/CAL Indeterminate	Add an INDETERMINATE state when <code> error </code> falls within U of the limit
No reference-adequacy check	CMX TUR view; MET/CAL	Call the existing <code>calculate_measurement_capability</code> ; warn below 4:1
Only two Type B components	CMX's eight user-defined	Make Type B a related table, not two fixed fields, required anyway for sensitivity coefficients
Signature is a pasted image	Ansur / MET/TEAM / CMX 21 CFR 11	Ed25519 sign the certificate payload; keep the image as decoration only
Verification is a stub	All three	Look up the certificate, check the signature, return a truthful <code>valid</code>
Manual transcription of readings	Ansur plug-ins; Beamex MC6	Longer-term: ingest from analyzers that expose a serial/USB interface
Interval by fixed period	MET/TRACK's use cycles	<code>estimate_calibration_interval</code> already exists (dead); drift-based intervals are a differentiator, not a catch-up

5. What CalSoft should deliberately *not* copy

Worth stating, because the temptation after a comparison like this is to chase every row.

Do not chase full MET/CAL procedure automation. MET/CAL's value is a scripting language that drives calibrators through thousands of procedures in an accredited electrical metrology lab. A hospital biomedical workshop calibrating NIBP modules, infusion pumps and defibrillators against handheld analyzers has a different problem. Automating *capture* from the analyzers is worth doing; reimplementing a procedure scripting engine is not.

Do not pursue 21 CFR Part 11 as a checklist. It governs FDA-regulated electronic records. Unless the hospital is in that regime, the parts to adopt are the *engineering* ones, real electronic signatures, an audit trail over the record, tamper-evident certificates, not the compliance apparatus.

Do not add uncertainty components for their own sake. CMX's eight Type B slots are useful because process instrumentation has many influence quantities. CalSoft's gap is not the *count* of components. It is that every component is assumed to have a sensitivity coefficient of 1, so no influence quantity needing a coefficient can enter at all (A1_ENGINEERING_AUDIT.pdf section 8). Fix the model, then the count follows.

Do not drop offline-first to match a commercial architecture. It is the one place CalSoft is ahead of all three.

6. Two realistic targets

Target A, "Ansur-plus" (weeks). Match the hospital-QA peer on records integrity while keeping the uncertainty advantage. Real e-signatures, working certificate verification, the verdict bug fixed, guard banding and TUR wired in from the code that already exists. On the section 1 matrix this takes CalSoft past Ansur on every row except automated capture and multi-device batch testing.

Target B, "CMX-class uncertainty" (months). Type B components as a related table with sensitivity coefficients, Welch-Satterthwaite effective degrees of freedom replacing the return $\max(2, 10)$ stub, a stated decision rule printed on the certificate, and drift-based interval estimation. This is where CalSoft would have a defensible ISO/IEC 17025 story.

Target A is almost entirely wiring code that has already been written and documented. Target B is genuine engineering. Nothing here requires research.

7. Verdict

Against **Fluke Biomedical Ansur**, its true functional peer, CalSoft is

ahead on measurement mathematics (Ansur's datasheet documents limit-based pass/fail and no uncertainty budget), **ahead on offline operation and cost** and **behind on automated data capture, multi-device testing, electronic signatures and record integrity**.

Against **Fluke MET/CAL + MET/TEAM** and **Beamex CMX**, CalSoft now has both the uncertainty engine and the decision machinery: guard banding with three outcomes, a declared decision rule, TUR warnings, and a derived coverage factor. What still separates it from them is **not metrology but records**: neither the certificate nor the signature is tamper-evident, the QR verifier still returns valid for any input, and there is no configurable per-capability decision rule.

The remaining honest gap is therefore narrower and more specific than it was: CalSoft computes and uses uncertainty about as well as the commercial products for the parameters a hospital workshop calibrates, and trails them on

automated data capture and on **proving a certificate is genuine**.

The summary as it now stands: **CalSoft computes uncertainty more thoroughly than the product sold for its job, and now reaches its verdict accordingly**. The decision-logic gap that defined this comparison is closed. What remains is the promise of QR verification that the code still does not keep, a records problem rather than a measurement one, and the one place where all three commercial products are clearly ahead.

8. Sources

Vendor documentation consulted 17 September 2026.

- Fluke Biomedical, *Ansur Test Automation Software, Technical Data* (ansur-3.0-datasheet_en.pdf): key features, plug-in list, electronic signature, MS SQL/.NET requirements
- Fluke, *MET/CAL Calibration Management Software* product pages and *New features in recent MET/CAL software versions*: GUM-compliant uncertainty since v6.0 (1999); guard banding yielding Pass / Fail / Indeterminate
- Fluke, *MET/TEAM* and *MET/TRACK Asset Management Software*: ISO/IEC 17025, ANSI Z540, 21 CFR Part 11 support; electronic signatures and audit trail; intervals in days, weeks, months or use cycles
- Beamex, *CMX Calibration Management Software* product pages and *CMX User Manual v2.17.1*: combined expanded uncertainty per calibration point; uncertainty page showing pass-with-risk-of-fail; TUR; up to eight user-defined Type B uncertainties; ISO 17025 / GMP / 21 CFR Part 11 posture
- CalSoft claims are cited to file and line in the working tree throughout

Vendor capabilities are stated as their own documentation describes them and were not independently tested. "Not documented" records the absence of a claim in that vendor's literature, not a proven absence of capability. CalSoft

claims are from direct source reading; no application code was modified.

